fn sample_geometric_exp_fast

Michael Shoemate

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This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of `fn sample_geometric_exp_fast` in `mod.rs` at commit 0be3ab3e6 (outdated[1]). This proof is an adaptation of subsection 5.2 of [CKS20].

1 Vetting history

- Pull Request #519

2 Pseudocode

Precondition

$\mathtt{x} \in \mathbb{Q} \wedge \mathtt{x} > 0$

Implementation

```
def sample_geometric_exp_fast(x) -> int:
    if x == 0:
        return 0

    s, t = into_numer_denom(x)

    while True:
        u = sample_uniform_int_below(t)
        d = sample_bernoulli_exp(Rational(u, t))
        if d:
            break

    v = sample_geometric_exp_slow(1)
    z = u + t * v
    return z // s
```

Postcondition

`sample_geometric_exp_fast` returns an integer sample from $Geometric(1 - exp(-x))$, or an error propagated from `sample_uniform_int_below` or `sample_geometric_exp_slow`.

[1] See new changes with `git diff 0be3ab3e6..cb54ec2db rust/src/traits/samplers/cks20/mod.rs`

3 Proof

Since x is a rational, let $x = s/t$.

1. Let u be a realization of a random variable $U \sim Uniform(0, t)$.
2. Let d be a realization of a random variable $D \sim Bernoulli(exp(-u/t))$
3. Let v be a realization of a random variable $V \sim Geometric(1 - exp(-1))$

Lemma 3.1. Conditioned on $d = \top$, if $z = u + t * v$, then z is a realization of a random variable $Z \sim Geometric(1 - exp(-1/t))$. Equivalently, $P[Z = z|D = 1] = (1 - e^{-(1/s)})e^{-(z/s)}$ [CKS20].

Proof. For any z, define $u_z := z \ mod \ t$ and $v_z := \lfloor z/t \rfloor$, so that $z = u_z + t \ v_z$.

$$\begin{aligned}
P[Z = z|D = \top] &= P[U = u_z, V = v_z|D = \top] \\
&= P[U = u_z|D = \top]P[V = v_z] \\
&= \frac{P[U = u_z]}{P[D = \top]} P[D = \top|U = u_z] * (1 - e^{-1})e^{-v_z} \\
&= \frac{1/t}{1/t \sum_{k=0}^{t-1} e^{-k/t}} e^{-u_z/t} * (1 - e^{-1})e^{-v_z} \\
&= (1 - e^{-1/t})e^{-u_z/t+v_z} \\
&= (1 - e^{-1/t})e^{-x/t}
\end{aligned}$$

□

Lemma 3.2. Fix $p \in (0, 1]$. Let G be a $Geometric(1 - p)$ random variable, and $n \geq 1$ be an integer. Then $\lfloor G/n \rfloor$ is a $Geometric(1 - q)$ random variable with $q = p^n$ [CKS20].

Proof.

$$\begin{aligned}
P[\lfloor G/n \rfloor = k] &= P[nk < G < (k + 1)n] \\
&= \sum_{l=kn}^{(k+1)n-1} (1 - p)p^l \\
&= (1 - p^n)p^{nk} \\
&= (1 - q)q^k
\end{aligned}$$

□

Theorem 3.3. Given any $s, t \in \mathbb{Z}_+$ and $Z \sim Geometric(1 - exp(-1/t))$, define $Y = \lfloor Z/s \rfloor$. Then $Y \sim Geometric(1 - exp(-s/t))$[CKS20].

Proof.

$$\begin{aligned}
P[Y = y|D = \top] &= P[\lfloor Z/s \rfloor = y|D = \top] \\
&= (1 - p^s)p^{sk} && \text{by 3.2} \\
&= (1 - (e^{-1/t})^s)(e^{-1/t})^{sk} \\
&= (1 - e^{-s/t})(e^{-s/t})^k
\end{aligned}$$

□

By 3.1, `z` is a realization of $Z \sim Geometric(1-exp(-1/t)$. Let y be the return value of `sample_geometric_exp_fast(x)`, assuming the preconditions are met, and the function does not return an error. Since z is a realization of $Z \sim Geometric(1 - exp(-1/t))$, then by 3.3, $y \sim Geometric(exp(-s/t))$.

References

[CKS20] Clément L. Canonne, Gautam Kamath, and Thomas Steinke. The discrete gaussian for differential privacy. *CoRR*, abs/2004.00010, 2020.